

from the internet, confusion between these terms will leave you in a whole lot of mess and you'll only make things worse.

## Uses in the world today

While virtualization is fun to play with, there are a lot of more uses than just testing little applications and trying out operating systems for fun. There are a lot of other uses which are not known to the general users as such. One of those is the great wor(l)d of cloud computing.

Cloud computing, is a way to make use of more than one computer to serve one purpose. This “one purpose” may consists of many other tasks and the “purpose” itself is to provide the software as service. When we say “software as a service” (or SAAS, as many would call it), we actually mean – a way of allowing the user to use the software without him allowing to install the software on his machine. One good way of understanding it is a “website”. Although not “truly” a SAAS, a website allows you to store your data on it and process it, search for it or utilize it for other things, but it does not need you to install any servers on your machine. Taking “Facebook” as an example, it allows you to upload your pictures, organize them and use them for getting comments and likes from your friends but it does not ask you to install the software it uses to facilitate all that.

Normally a small website such as a Blog would be able to run on a single machine which has the database, the web server and other components installed on it but when the data gets too large, it has to be distributed across many, many machines. Taking “Facebook” again as an example, it hosts more than 100 billion pictures. There is no way you could keep all of that on a single machine. In such cases the pictures are to be distributed across many machines connected to each other via a network. As the amount of data increases, it becomes cumbersome to manage all the data. With new machines being added all the time into the network with different (and improved) features and capabilities, the scene gets even messier.

Virtualization in such cases serves more than one purpose. A virtual environment allows the service provider to virtualize the storage and put the same software across all the machines without trouble. It increases the security and manageability of the whole system and with advanced virtualization software, interoperability of the machines, fault detection, dynamic resource allocation and load balancing can be done more easily. All that helps you visualize a system with large amounts of memory and storage available to you without hassles. It helps manage the network and other resources.